REVISION NOTES

Power Platform — Full Topic Set

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Digital Transformation: Tools and Techniques | Final Exam Prep

Modules Covered:

Initial: Power Apps with Model Driven Apps
Module 1: Create a Canvas App in Power Apps
Module 2: Get Started with Power Automate
Module 3: Administer Flows (Power Automate Admin Center)
Module 4: Create an Agent with Microsoft Copilot Studio
Module 5: Business Value of Microsoft Power Platform

Model-Driven Apps in Power Apps

Digital Transformation: Tools and Techniques

Final Exam Revision Notes

1. What Are Model-Driven Apps?

Model-driven apps are built in Power Apps using a **data-first approach**. Instead of designing the layout manually (like canvas apps), the layout is **determined by the data components** you add — forms, views, dashboards, and charts.

Key Characteristics

- Little to no code required
- Layout is determined by components, not the designer
- Data is stored and managed in Microsoft Dataverse
- Connects Dataverse tables via relationships to reduce data repetition
- Focus is on viewing business data and making decisions quickly

Canvas Apps vs. Model-Driven Apps

Feature	Canvas Apps	Model-Driven Apps
Layout control	Full control by designer	Determined by components
Code required	Low-code with formulas	Little to no code
Data source	Multiple connectors	Primarily Dataverse
Best for	Custom UI experiences	Data-heavy business apps

2. The 5 Phases of Building a Model-Driven App

1. Model your business data — Most important step. Build your data model in Dataverse first. Define what data you need and how it relates to other data.
2. Define your business processes — Create consistent workflows so users follow the same steps every time (e.g., approvals, service requests).
3. Compose the app — Use the App Designer to select and set up pages. Power Apps auto-creates a default site map for navigation.
4. Configure security roles — Assign roles to control who can access which Dataverse tables. Security applies across all apps using that data.
5. Share the app — Two-step process: (1) assign security role to user for Dataverse access, then (2) share the app itself.

> **Exam Tip:** *Sharing an app without assigning a security role means the user cannot use the app.*

3. Components of Model-Driven Apps

Components fall into 4 categories: Data, UI, Logic, and Visualization.

A. Data Components

Component	What It Is
Table	Container for records (like an Excel sheet). e.g., Contact, Account
Column	A property/field in a table. Each has a data type: text, number, date, currency, lookup
Relationship	Defines how tables relate: 1:N, N:1, or N:N
Choice Column	Gives users predetermined options. Two kinds: single-select and multi-select

Relationship Types

- 1:N (One-to-Many): One customer can have many quotes
- N:1 (Many-to-One): Many quotes belong to one customer
- N:N (Many-to-Many): Many students can attend many classes
- Adding a Lookup column creates a 1:N relationship between two tables

B. UI (User Interface) Components

Component	What It Does	Designer
App	Fundamental settings, properties, URL	App designer
Site Map	Navigation structure for the app	Site map designer
Form	Data-entry interface for a table (populates rows/columns)	Form designer
View	Defines how a list of records is displayed (columns, sort, filters)	View designer
Custom Page	Canvas-based page for flexible layouts	Canvas designer

C. Logic Components

Type	What It Does
Business Process Flow	Step-by-step guide that walks users through a process (Model-Driven apps only)
Workflow	Automates processes — triggered manually, on a schedule, or by events
Actions	Manually invoked actions from workflow, plugin, or button
Business Rule	Applies logic to a form: require fields, hide/show columns, validate data

Type	What It Does
Power Automate Flow	Cloud-based automation across multiple apps and services (broader than Workflows)

Business Rules vs. Business Process Flows

- Business Rules = field-level logic on a form (e.g., make a field required, show/hide a column)
- Business Process Flows = guided multi-step experience across the whole app

D. Visualization Components

Component	What It Does
Chart	Bar, pie, or other graphic displayed in views, forms, or dashboards
Dashboard	Collection of charts and visualizations for a business overview
Embedded Power BI	Power BI tiles/dashboards embedded directly in the app

4. Designing a Model-Driven App

Four key design considerations: Business Requirements, Data Model, Business Logic, and Output.

Business Requirements

- Understand security, accessibility, data, and design needs with stakeholders
- Consider: data security model, government regulations, authentication (e.g., MFA)
- Offline Mode: Supported on iOS and Android clients — requires additional design planning

Data Model

The three most critical elements: **Table → Column → Relationship**

- Model-driven apps use metadata-driven architecture — app design is based on data structure, no custom code needed
- Metadata = 'data about data' — defines the data structure stored in the system
- Column types affect user experience: Choice columns appear as dropdowns; Currency columns show currency symbols

> **Exam Tip:** *If you need to change a column type, you must delete and recreate it — all data in that column will be lost!*

Business Logic (Two Approaches)

- Business Rules — set field requirements, default values, show/hide fields (e.g., 'mileage' field only required if travel type = automobile)
- Business Process Flows — guided step-by-step experience; can require approvals between steps

Output (Dashboards)

- Create dashboards with filters and visual graphics
- Keep dashboards simple — high-level snapshots, allow users to drill down

Industry Accelerators

Microsoft provides pre-built solutions for specific industries: Health, Finance, Banking, Education, Non-Profit, Automotive, and Media.

5. Security Roles

Core Concepts

- Every Dataverse table must have a defined security role for users to access it
- A user can have multiple security roles, but must have at least one
- Roles can be assigned to individual users or teams
- To share an app, you must have Environment Admin or System Admin role
- Predefined security roles only work on standard tables — custom tables require custom security roles

Privilege Types — Memorize These!

Privilege	What It Allows
Create	Make a new record
Read	Open and view an existing record
Write	Make changes to a record
Delete	Permanently remove a record
Append	Associate the current record with another record (e.g., attach a note to an opportunity)
Append To	Associate another record with the current record (e.g., add a note to the current record)
Assign	Give ownership of a record to another user
Share	Give another user access to a record while keeping your own access

Lab Scenario — Pet Grooming App (Contoso)

Role	Privileges
Pet Grooming Technicians	Read, Write, Append (on Pet table) + Read (on Account table)
Pet Grooming Schedulers	Create, Read, Write, Delete, Append, Append To, Assign, Share (on Pet + Account tables)

Steps to Share a Model-Driven App

6. Sign into Power Platform admin center
7. Go to Settings > Users + Permissions > Security Roles
8. Create a new security role and set privilege levels per table
9. Assign users to the role (Members > Add People)
10. Go to make.powerapps.com > Apps > Select app > Share
11. Add users/groups and assign security role > Share

6. Business Process Flows (BPFs)

What They Are

- Online, step-by-step guided processes for users
- Only available in Model-Driven Apps
- Reduce training time — users don't need to know which table to use
- Visually show users where they are in a process

Key Facts

- Can span up to 5 different tables in a single process
- Up to 10 active BPFs per table
- Can be tailored to users with different security roles

Built-in / System BPFs (Examples)

- Lead to Opportunity Sales Process
- Opportunity Sales Process
- Phone to Case Process

When to Use BPFs

Use BPFs when you need:

- Consistent, guided steps (e.g., all staff handle requests the same way)
- Approvals between stages
- Multi-table workflows (e.g., Opportunity > Quote > Order > Invoice)

7. Key Exam Reminders

- Model-driven apps are data model driven — the data structure comes first
- The App Designer is used to compose the app (not build it from scratch with code)
- Power Apps auto-creates the default site map (navigation)
- Dataverse is the underlying data platform for model-driven apps
- Sharing requires both: assigning a security role AND sharing the app
- Business Rules = form-level logic | Business Process Flows = app-level guided experience
- Power Automate Flows work across multiple apps; Workflows are Model-Driven app-specific
- Metadata-driven architecture = no code needed to change app design when data model changes
- Lookup column = creates a 1:N relationship between two tables
- Changing a column type requires deleting and recreating it — data loss warning!

> **Exam Tip:** *The lab scenario (Pet Grooming / Contoso) is highly likely to appear in conceptual exam questions — know the two roles and their privilege differences.*

MODULE 1: Create a Canvas App in Power Apps

> *Learning Path: 6 Modules*

1.1 What is Power Apps?

Power Apps is a suite of apps, services, and connectors that lets you build custom business apps **without coding**. It provides a rapid development environment to solve business problems across finance, sales, HR, and operations.

Key Capabilities:

- Build apps from existing data sources (Excel, SharePoint, Dataverse, SQL)
- Works in web browsers and on mobile devices
- Share apps instantly with coworkers on phones and tablets
- Uses **AI-powered development (Copilot)**, prebuilt templates, and 1,400+ connectors

Real-World Examples:

- Field reps use a tablet app to view customer equipment, test parts, and order replacements on-site
- Restaurant staff use a phone app to record shifts and request vacation instead of paper

1.2 The Power Apps Maker Portal

Access at: **make.powerapps.com**

Key Areas:

- **Create** — start a new app (from Copilot, data, page design, or blank canvas)
- **Apps** — list of your existing apps
- **Home** — learning resources, templates
- **Do more with your apps** — Copilot agents, Power Automate flows, websites, AI Hub

> *Apps created with Copilot use Microsoft Dataverse as the data source.*

Power Apps Mobile — free app for iOS and Android to run canvas apps.

1.3 Data Sources

Data Source	Cost	Best For
SharePoint	Free	Lists as backend; small-medium data
Excel	Free	Learning, small datasets
Dataverse	Premium	Large datasets, relationships, Copilot features
SQL	Premium	Enterprise-grade, large-scale apps

SharePoint Key Rules:

- Use simple column types (Text, Number, Yes/No, Date)
- Avoid mandatory columns — enforce requirements inside the app instead
- No relational table support — create key fields manually for relationships
- Subject to **delegation limit** (queries returning too much data show a warning)

Excel Key Rules:

- Data MUST be formatted as a Table in Excel
- Image columns must be labeled `[image]`
- Not recommended for multi-user live apps (file locks)

Dataverse Key Benefits:

- No API configuration needed
- Supports large datasets, automatic table relationships
- Fully supports Copilot for natural language app generation

1.4 Power Apps + Power Automate + Power BI

Power Automate

- Creates automated workflows triggered by Power Apps events
- Can: update data sources, download data, generate PDFs, send email, start approvals, post to Teams
- **Power Automate Desktop** automates legacy desktop systems (RPA — Robotic Process Automation)

> *When to use Power Automate: when business logic exceeds what Power Apps handles natively (e.g., multi-step approvals, scheduled flows)*

Power BI

- Analytics and reporting tool; connects to multiple data sources
- Creates interactive reports and dashboards
- Sharing requires **Power BI Pro license** for both creator and viewer

Integration Options:

- Embed a Power BI tile **inside** a Power Apps canvas app
- Embed a Power Apps canvas app **inside** a Power BI dashboard

> *When to use Power BI: advanced analytics, KPI dashboards, trend analysis — not for simple charts in an app*

1.5 Designing a Canvas App

Before building, answer:

1. What should the app do?
2. Are we replacing an analog process?
3. Is mobile required?
4. How many data rows will it handle?

Four Design Considerations:

5. **Business Requirements** — Security, privacy, compliance, authentication (e.g., MFA), offline needs
6. **Data Model** — Choose data source based on existing infrastructure, volume, licensing cost
7. **UX (User Experience)** — Keep it simple and intuitive; avoid high-res images that slow mobile performance; use sliders instead of manual input where possible; plan for accessibility and localization
8. **UI (User Interface)** — Create mockups first (Visio, PowerPoint, or paper); test layout before building

1.6 Customizing a Canvas App

Gallery Control

- Displays a list of records from a data source
- Select the gallery → change **Layout** in Properties panel (e.g., "Image, title, subtitle, and body")
- Each field shows `ThisItem.ColumnName` in the formula bar
- Modify fields by selecting gallery → **Edit fields** in command bar

Form Control

- Shows/edits details of a single record
- Key properties: **DataSource** (where data is written) and **Item** (what record is shown)

Inside a form, each field is a DataCard containing:

- `DataCardKey` — label
- `DataCardValue` — input control
- `StarVisible` — asterisk if field is required
- `ErrorMessage` — shown when required input is missing

- To save a form: button must have `OnSelect = SubmitForm(Form1)`
- **Default** property = column that pre-fills the card
- **Update** property = where input is written back to the data source

Controls in Canvas Apps

Control Type	Purpose
Gallery	Display a list of records
Form	View and edit a record
Media	Camera, barcode reader, images
Charts	Visual data including Power BI tiles

Formulas in Canvas Apps

- `ThisItem.ColumnName` — references data from the current gallery row
- `IsSelected` — Boolean; true for the selected gallery item
- `Select(Parent)` — default OnSelect for gallery controls; selects that item

1.7 Managing Apps

App Versions

- Power Apps saves version history in the cloud
- Users see only the published version
- Restoring a version creates a new version number (e.g., restoring v3 when v5 exists → creates v6)
- Versions are kept for **6 months**
- To make a restored version available: you must publish it

Sharing Apps

Two ways to share:

9. Inside Power Apps Studio → **Share button** (top right)
10. From Apps tab → select app → Share

Permission Levels:

Role	Permissions
Co-owner	Can use, edit, and share; cannot delete or change owner
User	Can only use the app

> ? *If the app uses Dataverse, users must also have correct security roles for the relevant tables.*

Environments

An **environment** is a container for apps, data connections, and flows.

Reasons to create separate environments:

- Separate departments
- Support ALM (development → testing → production)
- Manage data access independently

Type	Purpose
Developer	Personal use; no expiry; includes premium features
Production	Live apps and data; the Default environment is Production
Trial	Temporary exploration; expires after 30 days
Trial (subscription-based)	Multi-user; has an end date, extendable
Sandbox	Non-production Dataverse testing; isolated from production

Environment Roles:

- **System admin** — full permissions; can create/manage environments
- **Environment maker** — can create/modify apps and work with Dataverse

> *Creating an environment requires at least 1 GB of available tenant capacity.*

MODULE 2: Get Started with Power Automate

2.1 What is Power Automate?

Power Automate is an **online workflow service** that automates actions across apps and services. It connects to hundreds of services and can manage cloud and on-premises data.

Common Use Cases:

- Respond to high-priority emails instantly
- Copy email attachments to OneDrive automatically
- Capture and follow up with sales leads
- Automate approval workflows
- Monitor social media mentions
- Sync files between services (e.g., Dropbox → SharePoint)

2.2 Key Concepts: Triggers and Actions

Every flow has:

- **One trigger** — the event that starts the flow (e.g., new email, new list item)
- **One or more actions** — what happens when the trigger fires (e.g., create a file, send an email, post to Teams)

2.3 Types of Flows

Flow Type	Description
My flows	Your personal flows

Flow Type	Description
Templates	Prebuilt flows you can customize
Approvals	Manage approval processes and business process flows
Solutions	Containers that hold multiple components including flows
Process mining	Analyze workflows to find optimization opportunities

2.4 Creating Flows

Three main ways to create a flow:

11. **From a template** — quickest; just configure a few settings
12. **From a scheduled flow** — runs at set intervals (e.g., daily)
13. **From scratch** — full control; define your own trigger and actions

2.5 Sharing Flows

You can share flows directly from the **Power Automate mobile app** (Buttons tab):

- Select the flow → ellipsis → Invite others
- Search for user/group → Send

Managing Connections:

- Can allow shared users to use your connections or require them to use their own connections
- If using your connections, they cannot access your credentials or reuse connections in other flows

- **To stop sharing:** ellipsis → Invite others → select user → Remove user
- **Resharing:** Ellipsis → Share button link → send via any app

2.6 Troubleshooting Flows

Viewing Errors:

- Power Automate sends a weekly failure email by default
- View Cloud flow activity: left menu → More → Cloud flow activity

Error Code	Meaning	Fix
401 / 403 / "Unauthorized"	Authentication failure	Fix the connection → Resubmit
400 / "Bad request"	Action configuration issue	Edit flow → fix settings → Resubmit
404 / "Not found"	Action configuration issue	Edit flow → fix settings → Resubmit
500 / 502	Temporary/transient error	Resubmit

Common Limits:

- Max **600 flows** per account
- Max 50 custom connectors per account
- Max 100 connections total (20 per API)

- A gateway can only be installed in the default environment
- Free plan: flows run every 15 minutes minimum

> *▯ Microsoft accounts (@outlook.com, @gmail.com) can only use the free plan. Use an organizational account for paid features.*

MODULE 3: Administer Flows (Power Automate Admin Center)

3.1 The Admin Center

The **Admin Center** is where tenant and environment admins manage **data policies** and **environments**.

Two ways to open it:

14. Power Automate → Settings (gear) → **Admin Center**
15. Go directly to **admin.powerplatform.microsoft.com**

> *▯ Changes made in the Admin Center are immediately available to all users.*

3.2 Environments (Admin View)

An environment is a space to store, manage, and share business data, apps, and flows.

Common Environment Patterns:

- Separate test and production versions
- Department-specific environments
- Global branch environments
- All apps in a single environment

3.3 Data Policies

Power Automate **automatically uses your organization's existing security roles** — users cannot access data they don't already have access to. Organizations can optionally add data policies to **proactively block flows** that violate certain policies.

3.4 Exporting and Importing Flows

Exporting a Flow

16. My flows → Cloud flows → select flow → More commands (...) → Export → Package (.zip)
17. Fill in Name, Environment, Description, and Review Package Content
18. Select Export → saves a zip file

> *▯ Solutions are now the preferred method to move flows between environments. Zip export still works but is considered legacy.*

Importing a Flow

19. My flows → Import
20. Upload the zip file
21. Configure: create new flow or update existing; set required connections
22. Select Import

3.5 Sharing Button Flows

Done from the **Power Automate mobile app** (Buttons tab):

- Share → invite a user or group
- Choose whether they use your connections or their own
- Can stop sharing at any time by removing users
- Users can reshare via Share button link
- To use a shared button: Buttons tab → Get more → select and add
- To stop using: Buttons tab → ellipsis → Remove

> ▯ *Monitoring: The Activity tab in the mobile app shows the complete run history, including runs by people you shared with.*

MODULE 4: Create an Agent with Microsoft Copilot Studio

4.1 Overview

Microsoft Copilot Studio (formerly Power Virtual Agents) allows you to create **intelligent conversational agents** — no high-code solutions needed. Agents can be deployed in Microsoft Teams and answer employee questions by pulling data from **Dataverse for Teams**.

Benefits:

- No coding required
- Can pull data from Dataverse tables
- Can be published to a team or the whole organization
- Performance analytics available over time

4.2 Creating an Agent

To install Copilot Studio in Teams:

23. Open Teams → Apps → search "copilot" → install **Microsoft Copilot Studio**

To create a new agent:

24. Open Copilot Studio for Teams → Start now under "Create an agent"
25. Select the owning team
26. Provide **Agent name** and **Language** (language cannot be changed after creation)
27. Select Create

> ▯ *Team owners and creators have read/write access by default; team members have read access only.*

You can also create agents in **Microsoft 365 Business Chat** at microsoft365.com/chat → Copilot pane → Create an agent (no code required).

4.3 Topics

Topics define agent conversations. Two types of nodes:

Trigger nodes — phrases, keywords, and questions the user is likely to enter.

> ⍰ *Recommend 5–10 trigger phrases per topic*

Conversation Nodes:

Node Type	Description
Ask a question	Have agent ask a question and get user response
Add a condition	Add branching logic
Call an action	Call Power Automate flows or authenticate
Redirect to another topic	Send user to another topic
Show a message	Agent responds with text
End with survey	End conversation + satisfaction survey

System Topics (auto-generated):

- Greeting, Goodbye, Escalate, Start Over, End of Conversation (all customizable)

Creating a Topic

28. Topics page → + New topic → From blank
29. Details → name the topic
30. Add trigger phrases (e.g., `who are the event contacts`)
31. Add message nodes with response text
32. Add Redirect to another topic → End of Conversation at the end
33. Save

> ⍰ *Test using the Test panel (left side of authoring canvas).*

4.4 Inputs, Variables, and Conditions

- **Inputs** — the user's response in an Ask a question node
- **Variables** — store inputs to use in later conversation nodes (e.g., `VarCountry`)
- **Conditions** — branching logic based on variable values

For multiple-choice questions:

- Select Multiple choice options in the Identify field
- Copilot Studio creates a condition node for each option
- Variable is named when you edit **Save response as**

Built-in Variables:

- `bot.UserDisplayName` — the user's name
- `bot.UserID` — the user's sign-in name

> ⍰ *These can be inserted into message nodes using the {X} insert variable dropdown.*

4.5 Publishing and Sharing an Agent

Publish the agent:

34. Copilot Studio → Publish (left menu) → Publish button → confirm
35. Edit details: color, icon, short/long description, Teams app store info

Availability Options After Publishing:

- **Copy link** — share directly with users
- **Add to a team** — team members can chat with agent privately or @mention in channels
- **Show in Teams app store:**
  - **"Built by your colleagues"** — visible to teammates and shared users
  - **"Added by your org"** — visible to everyone (requires admin approval)

Publishing to the Whole Organization:

- Submit for admin approval
- After approval, appears in Added by your org section in Teams app store
- Any user can then discover and install it

> ? *Editing the agent after publishing requires republishing for changes to reach users.*

MODULE 5: Business Value of Microsoft Power Platform

> ? *Learning Path: 2 Modules*

5.1 The Four Core Products of Power Platform

Product	What It Does
Power Apps	Low-code app development for web and mobile; uses Dataverse
Power Automate	Automated workflows between apps and services
Power BI	Business analytics, reports, dashboards, data visualizations
Power Pages	Secure, low-code external-facing business websites using Dataverse data

5.2 Supporting Tools

Tool	What It Does
Microsoft Copilot Studio	Low-code tool to create custom AI agents; customizes Microsoft 365 Copilot
Connectors	Bridges linking apps, data, and devices; 1,400+ prebuilt connectors
AI Builder	Adds AI capabilities to workflows and Power Apps without writing code
Microsoft Dataverse	Scalable data platform; stores and manages data using a common data model
Power Fx	Low-code Excel-like programming language used across Power Platform
Managed Environments	Secure, isolated environments for building, testing, and deploying apps

5.3 Business Challenges Power Platform Addresses

Challenge	Detail
App demand outpacing delivery	Demand is often 5x greater than capacity
Evolving workforce expectations	Millennials/Gen Z expect tailored digital experiences
High cost of custom development	Development + ongoing maintenance is expensive
Need for agility	Traditional development takes too long; businesses need fast solutions
Scaling development	Need citizen developers (power users) alongside professional developers — fusion development

5.4 Connectors

Connectors are bridges that link different services and data sources within the Power Platform.

They work in:

- **Power Apps** — link apps to data sources
- **Power Automate** — act as triggers (start a process) or actions (perform a task)
- **Power Pages** — display live data on web pages
- **Power BI** — pull data from ERP systems or databases for dashboards

Real-World Example (Invoice Processing):

36. Outlook watches for incoming invoices
37. SharePoint saves attachments
38. Teams sends approval requests
39. Oracle adds approved invoices to ERP
40. Outlook sends confirmation emails to vendors

5.5 Microsoft Dataverse (Fundamentals View)

Dataverse is a **cloud-based relational database** platform that stores structured business data in **tables** (rows and columns).

Key Features:

- **Standard tables** for common scenarios (accounts, contacts, activities)
- **Custom tables** for unique organizational needs
- **Role-Based Access Control (RBAC)** — only authorized users can access data
- Field-level security and auditing for compliance
- Integrates with Azure, Dynamics 365, Microsoft 365, Salesforce, SAP
- Supports scheduled data sync, data transformation, and one-time imports

Dataverse in the Power Platform Ecosystem:

Product	Role of Dataverse
Power Apps	Backend for apps; supports offline sync on mobile
Power Automate	Connectors for automation; enforces data validation rules

Product	Role of Dataverse
Power BI	Integrates with managed data lake for analytics
Power Pages	Builds dynamic data-driven web portals

5.6 Copilot and Generative AI in Power Platform

Generative AI creates new outputs (text, images, etc.) based on patterns in large datasets — unlike traditional AI which only predicts/classifies.

Copilot is the AI assistant embedded across Power Platform. Users interact with it using **natural language** instead of writing code or navigating menus.

Benefits:

- Saves time by automating repetitive tasks
- Provides intelligent suggestions based on user input
- Accelerates prototyping and refinement

Tool	Copilot Capability
Power Apps	Describe app purpose → Copilot creates Dataverse tables and app components
Power Automate	Describe a workflow → Copilot generates the flow and suggests connectors
Power Pages	Describe a web page requirement → Copilot generates site structure and forms
Power BI	Describe the report/dashboard needed → Copilot creates visualizations and identifies trends

5.7 Power Platform + Microsoft 365 Integration

Popular Microsoft 365 Connectors:

- Office 365 Outlook, OneDrive, Excel, SharePoint

Practical Examples:

- **Outlook + Power Automate** — automatically save email attachments to SharePoint (saves ~2 hrs/week)
- **Excel + Power Automate Desktop** — automate data entry from legacy systems using RPA

Power Platform in Microsoft Teams:

Component	What It Does in Teams
Power BI	Share interactive reports directly in Teams channels and chats
Power Apps	Design, edit, and publish apps directly within Teams
Power Automate	Build flows triggered from Teams messages; use Adaptive Cards
Copilot Studio	Create and share conversational agents in Teams
Dataverse for Teams	Built-in low-code data platform for Teams apps, bots, and workflows

5.8 Microsoft Copilot Studio (Full Overview)

Copilot Studio enables creation of **custom AI agents** without advanced technical skills.

Key Features:

- **Simple, low-code interface** — describe what you want; Copilot handles the technical setup
- **Seamless data integration** — 1,400+ connectors to Microsoft tools and third-party apps
- **Automated workflows** — set up Power Automate workflows as instructions for the agent
- **Multichannel deployment** — website, mobile app, social media, Microsoft Teams
- **Robust security and governance** — data protection and compliance controls

Dynamics 365 Customer Engagement (CE) Trial Setup — you spin it up through trials.dynamics.com or the Microsoft 365 admin center, pick a CE app (Sales, Customer Service, etc.), and get a 30-day trial environment provisioned in Power Platform Admin Center.

What's Included in D365 CE — CE is the customer-facing suite of Dynamics 365. It covers Sales, Customer Service, Field Service, Marketing (now Customer Insights - Journeys), and Project Operations. Under the hood it runs on Dataverse, so everything you just studied in Power Platform applies directly.

Navigation — the Unified Interface is the current UI. You have the sitemap (left nav), app switcher, command bar, views, forms, and dashboards. It replaced the legacy web client a few years back.

Who Needs CE — any org managing customer relationships: sales teams tracking leads/opportunities, support teams managing cases, field service orgs dispatching technicians, marketing teams running journeys.

CRM Modules in D365 CE — Sales (leads, opportunities, quotes, orders), Customer Service (cases, queues, SLAs, knowledge base), Field Service (work orders, scheduling, IoT), Marketing/Customer Insights (segments, journeys, emails), and Project Operations (projects, resources, billing).

Terminology — things like Accounts, Contacts, Leads, Opportunities, Cases, Activities, Business Units, Security Roles, Teams, Queues, Views, Forms, Dashboards, Solutions, Publishers.

Modules vs Add-Ons — first-party modules are licensed apps (Sales Pro/Enterprise, Customer Service Pro/Enterprise). Add-ons extend them — things like Sales Premium (adds Relationship Intelligence, Copilot features), Customer Insights - Data (CDP), LinkedIn Sales Navigator integration, etc.

Configuration vs Customisation — configuration is what you do inside the UI without code (fields, views, forms, business rules, workflows, security roles). Customisation goes deeper — custom entities, plugins, PCF controls, web resources, custom APIs, and Power Automate flows. The general rule is configure first, customise only when configuration can't do it.

Use Cases:

Domain	Agent Use Case
Customer Support	Answer questions, track orders, guide returns
Employee Assistance	HR FAQs, onboarding, internal navigation

Domain	Agent Use Case
Sales & Marketing	Product info, lead generation
Education	Study resources, scheduling, assignments
Healthcare	Appointment booking, medical records access
Project Management	Deadline tracking, task assignment, status updates

KEY EXAM REMINDERS — All Topics

41. Canvas apps give full layout control; model-driven apps use data-driven layout
42. Power Apps Maker Portal is at make.powerapps.com
43. Apps created with Copilot require Dataverse as data source
44. SharePoint delegation limit = queries returning more data than allowed show a warning
45. Excel must be formatted as a Table for Power Apps to read it
46. SubmitForm(Form1) is required to save changes made in a form
47. Restoring an app version creates a new version number and does NOT auto-publish
48. App versions are retained for 6 months
49. Co-owner can use, edit, and share — cannot delete or change owner
50. Power Automate Trigger = starts the flow; Action = what happens next
51. Error 401/403 = authentication issue; Error 400/404 = config issue; Error 500/502 = temporary
52. Max 600 flows per account; versions retained per environment settings
53. Solutions are the preferred way to move flows between environments (not zip export)
54. Copilot Studio agent language cannot be changed after creation
55. Recommended 5–10 trigger phrases per topic
56. Publishing an agent to the whole org requires admin approval
57. After editing an agent, you must republish for users to see changes
58. Power Platform has 4 core products: Power Apps, Power Automate, Power BI, Power Pages
59. 1,400+ prebuilt connectors available across Power Platform
60. Fusion development = citizen developers + professional developers working together
61. Dataverse uses RBAC (Role-Based Access Control) for data security
62. Copilot uses natural language to generate apps, flows, reports, and web pages
63. Power Fx = Excel-like low-code language used across Power Platform

End of Revision Notes • Good luck on your exam!